

Maryam Mirzakhani

School of Mathematics and Science

COHORT 1 | 23, 24, 25 May, 2026

This curriculum is inspired by Maryam Mirzakhani's playful, visual, and narrative approach to STEM education. Designed for primary students (Grade 5 and below) preparing for the Bangladesh Mathematics and Science Olympiad (BdMSO) 2026, this syllabus prioritizes deep conceptual intuition, spatial reasoning, and creative problem-solving over rote memorization.

Pedagogical Principles

1. Every student will receive a sketchbook. Complex math equations and scientific systems are drawn out first, focusing on visual structures rather than memorizing symbols.
2. We prioritize why a math system works (e.g., modular arithmetic clocks) over speedy mental arithmetic drills.
3. Science concepts (like genetic traits or the carbon cycle) are framed as natural mysteries, positioning the student as an active detective.
4. The material focuses heavily on the structural principles tested in BdMSO (pigeonhole principle, combinations, Delta geography, basic genetics, and force vectors) presented in an intuitive, age-appropriate manner.

Part 1: Mathematics Syllabus & Routine

Total Math Sessions: 6 Sessions | Total Hours: 10.5 Hours

Date & Time	Theme & Key Topics
Day 1 23 May 2026 Session 1 2.00 PM - 3.30 PM	The Language of Numbers Prime & composite numbers, Prime factorization, Number systems
Day 1 23 May 2026 Session 2 4.00 PM - 6.00 PM	Chasing Patterns & Remainder Clocks Divisibility rules, Remainder patterns & modular arithmetic, Arithmetic/geometric sequences
Day 2 24 May 2026 Session 3 9.00 AM - 10.30 AM	Symmetry, Mirrors, and Flatland Properties of 2D shapes, Transformations, Angle relationships
Day 2 24 May 2026 Session 4 11.00 AM - 1.00 PM	Architects of the 3D World Properties of solid figures, Nets of cubes and parallelepipeds, Surface area & volume
Day 3 25 May 2026 Session 5 9.00 AM - 10.30 AM	The Art of Counting Without Counting Systematic counting & ordered pairs, Basic combinatorics techniques, Permutations & combinations, Pigeonhole Principle Visual probability
Day 3 25 May 2026 Session 6 11.00 AM - 1.00 PM	The Final Maze: Logic & Recreational Math Logic grids & deductive puzzles, cryptarithms & magic squares, working backward

Part 2: Science Syllabus & Routine

Total Science Sessions: 6 Sessions | Total Hours: 10.5 Hours

Date & Time	Theme & Key Topics
Day 1 23 May 2026 Session 1 9.00 AM - 10.30 AM	The scientific method, Units, scale, estimation, Data interpretation & graphing.
Day 1 23 May 2026 Session 2 11.00 AM - 1.00 PM	Living vs non-living things, Cells and Cell organelles, Classifying organisms. Photosynthesis, Respiration and Transpiration.
Day 2 24 May 2026 Session 1 2.00 PM - 3.30 PM	Geography of Bangladesh and Bengal Delta formation. Climate change & environmental issues, Natural disasters. Ecology and human-environment interactions. Ecosystems, habitats, food webs and food chains.
Day 2 24 May 2026 Session 2 4.00 PM - 6.00 PM	Forces and effect of forces. Energy, sources of energy and energy conversions. Earth rotation & moon phases, Eclipses. Properties of Light and Sound waves.
Day 3 25 May 2026 Session 1 2.00 PM - 3.30 PM	Atoms & molecules, states and transformation of matter. Thermal energy, temperature and heat transfer. Applications of Electricity and Models of Magnetism.
Day 3 25 May 2026 Session 2 4.00 PM - 6.00 PM	Human Organ systems & skeletal structure, Nervous system. Nutrition, Common diseases and prevention. Genetics and biotechnology. GMOs, Biofuels.